Lake Powell and Lake Mead Historical and Current Reservoir Data

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Overview

This is an R Markdown document. The document generates plots for Lake Powell and Lake Mead:

1. Annual Lake Powell Release Compared to 7.5 and 8.23 Targets

Data wrangling strategy

1. Import the reservoir elevation-volume curves.
2. Import Daily, Monthly, and Annual Lake Powell and Lake Mead data from the Reclamation's HydroData Web portal https://www.usbr.gov/uc/water/hydrodata/reservoir_data/site_map.html
3. Filter the reservoir data for the appropriate variables/fields.

This is a beginning R-programming effort! There could be lurking bugs or basic coding errors that I am not even aware of. Please report bugs/feedback to me (contact info below)

Requested Citation

David E. Rosenberg (2026), "Lake Powell and Lake Mead Historical and Current Reservoir Data." Utah State University. Logan, Utah. https://github.com/dzeke/ColoradoRiverCollaborate/tree/main/HistoricalCurrentReservoirData.

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Figure 1: Annual Lake Powell Release Compared to 7.5 and 8.23 Targets

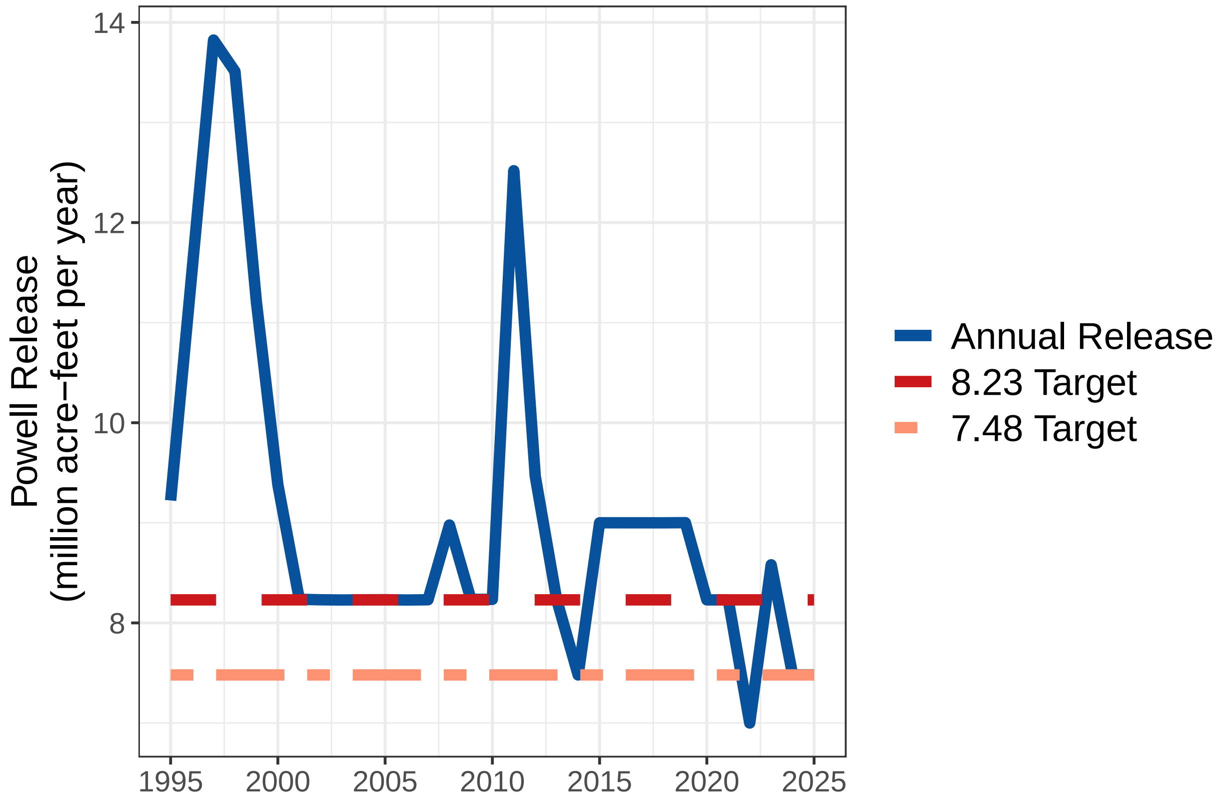

Figure 1: Ten-Year Powell Release Compared to 10-year Targets

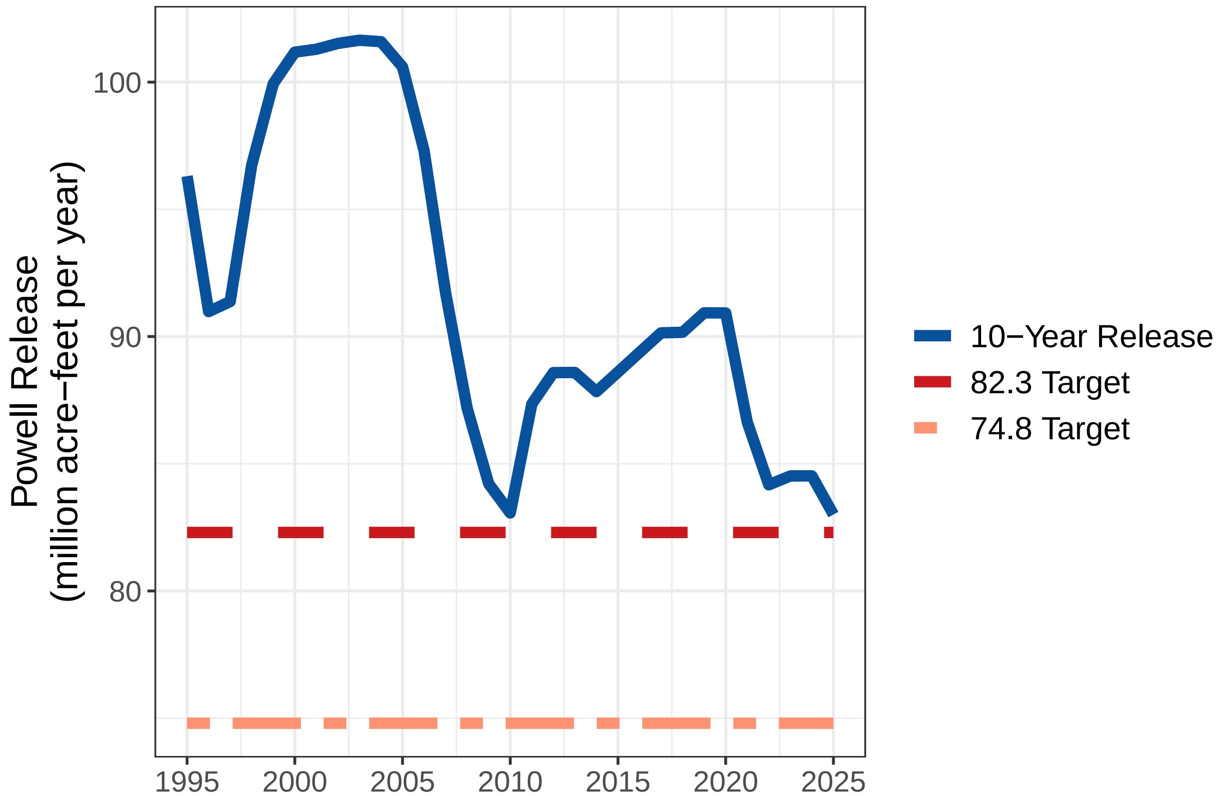

Table 1: Current Storage and Elevations

```
[1] "Reservoir Data on: June 5 , 2026"

# A tibble: 2 x 3
# Groups:   ResName [2]
  ResName      Storage `Pool Elevation`
  <chr>          <dbl>            <dbl>
1 Lake Powell     5.71            3528.
2 Lake Mead       7.60            1049.
```

Figure 3: Lake Powell, Lake Mead, and Combined Storage

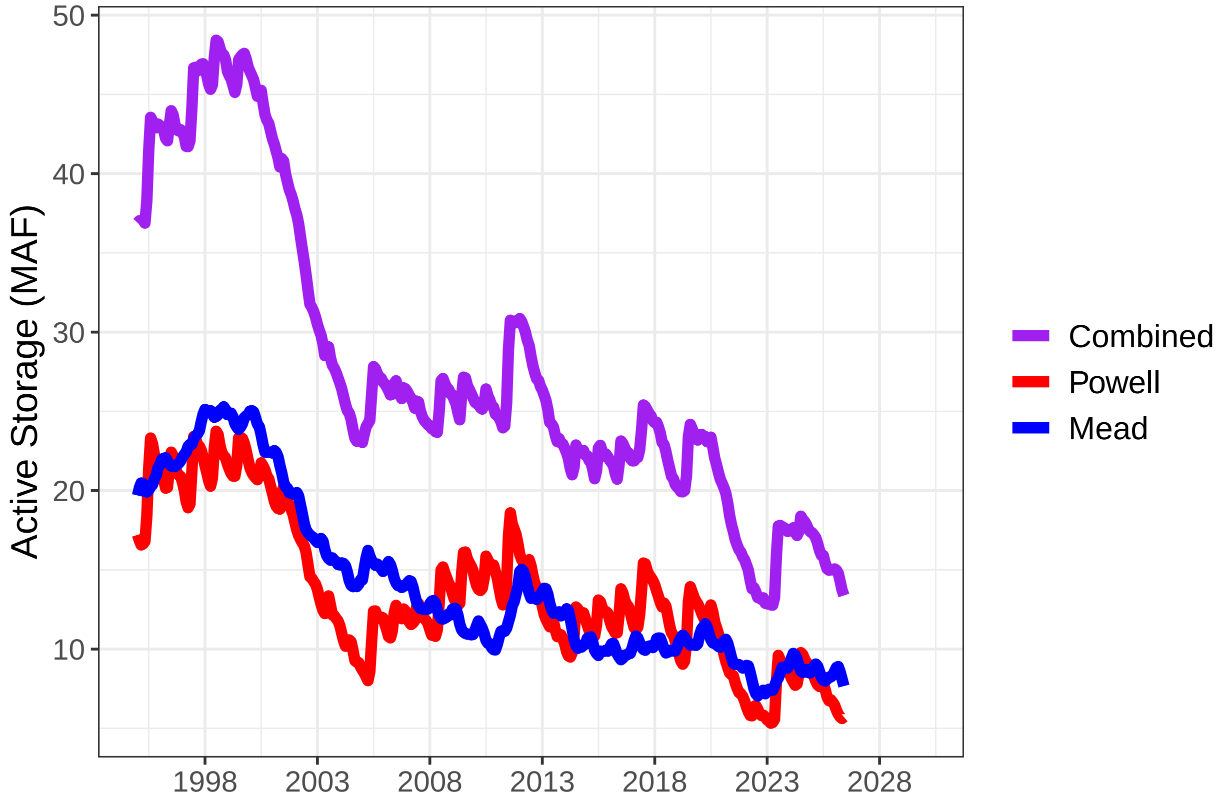

Figure 4: Lake Powell Storage vs Lake Mead Storage

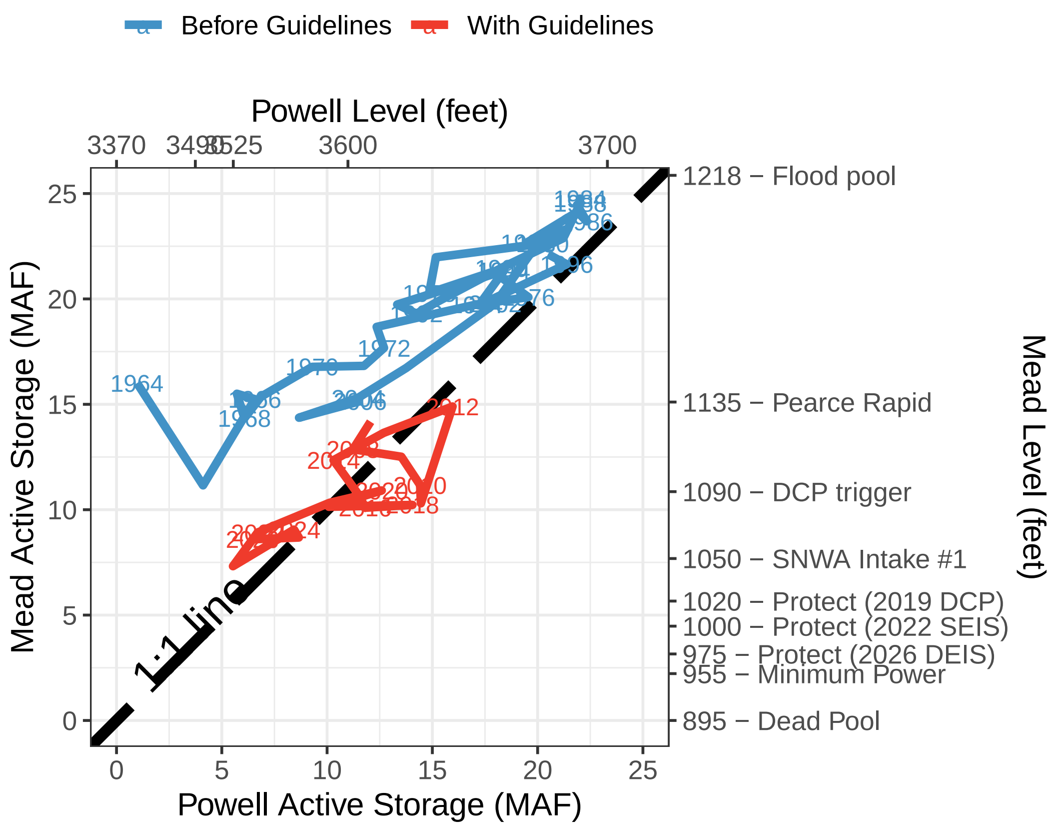

Figure 5: Lake Powell Storage and Elevation over Time.

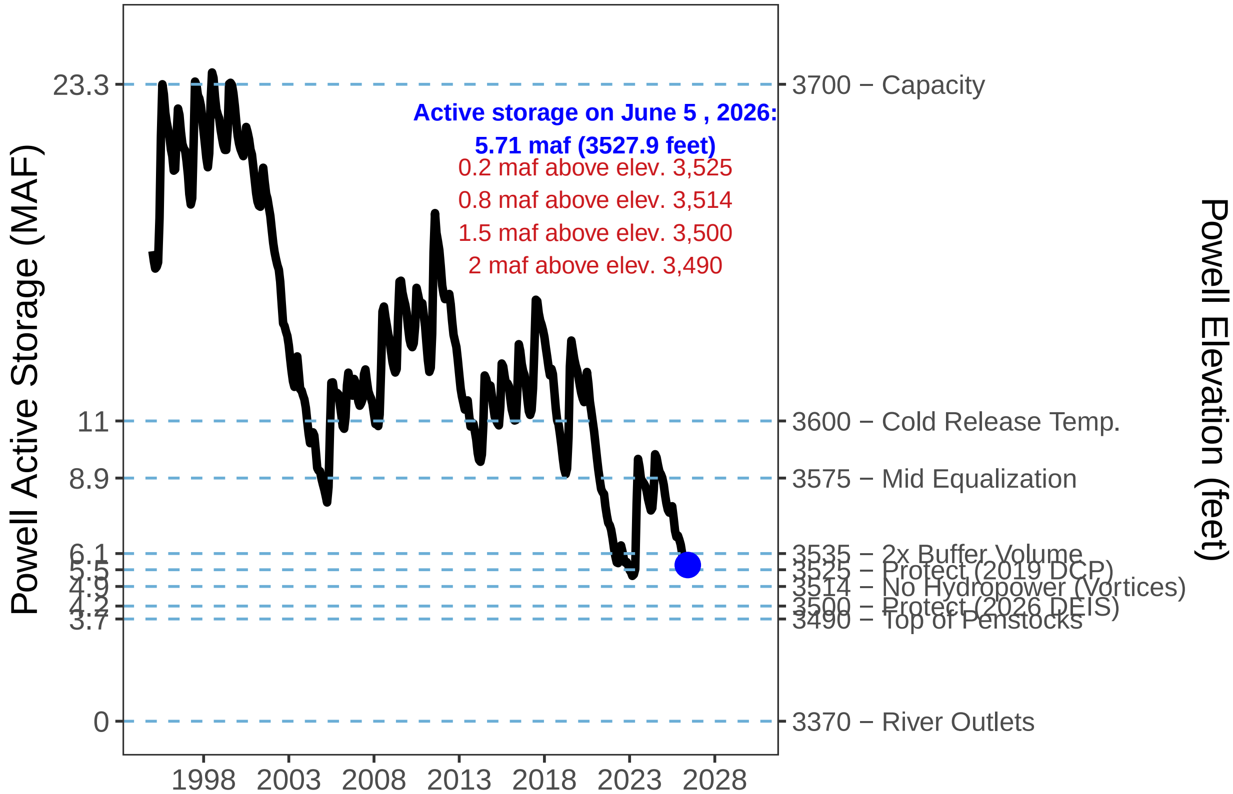

Figure 5B: Lake Powell Minimum Drawdown and Catastrophic Elevations

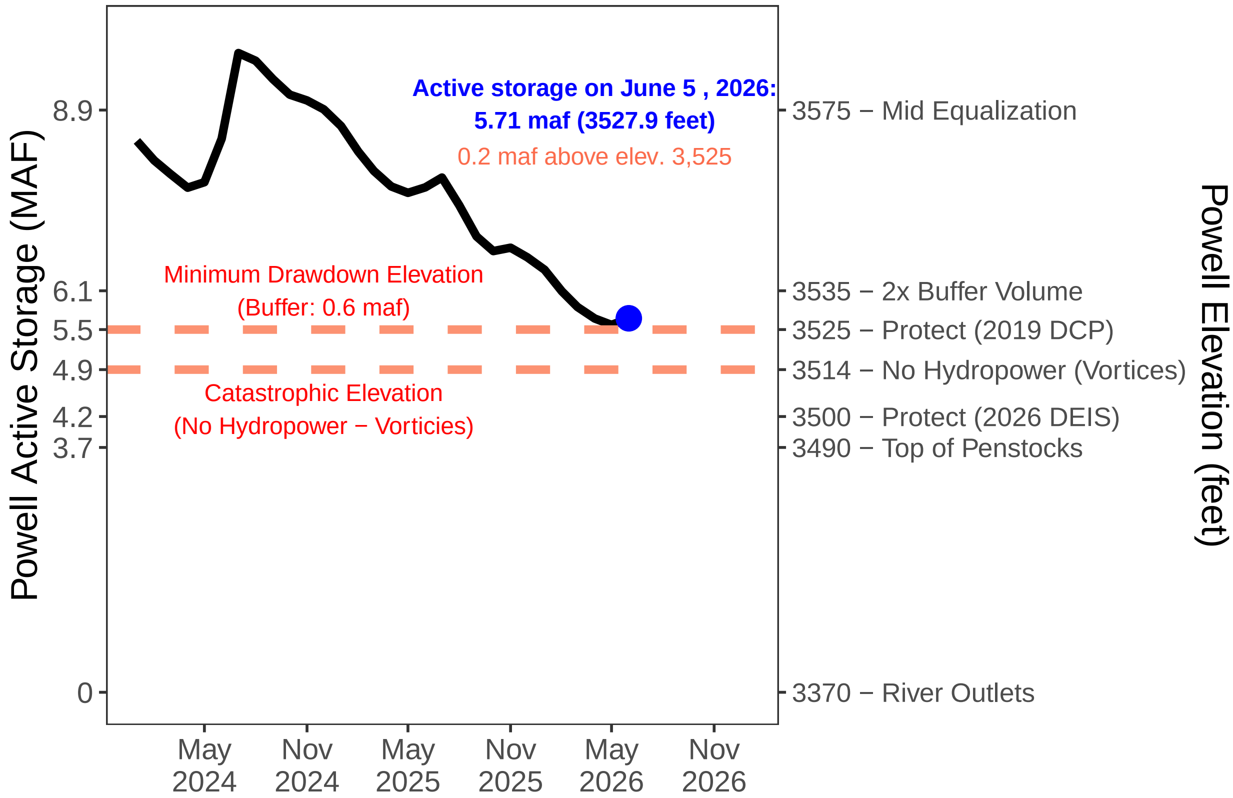

Figure 6: Lake Mead Storage and Elevation over Time.

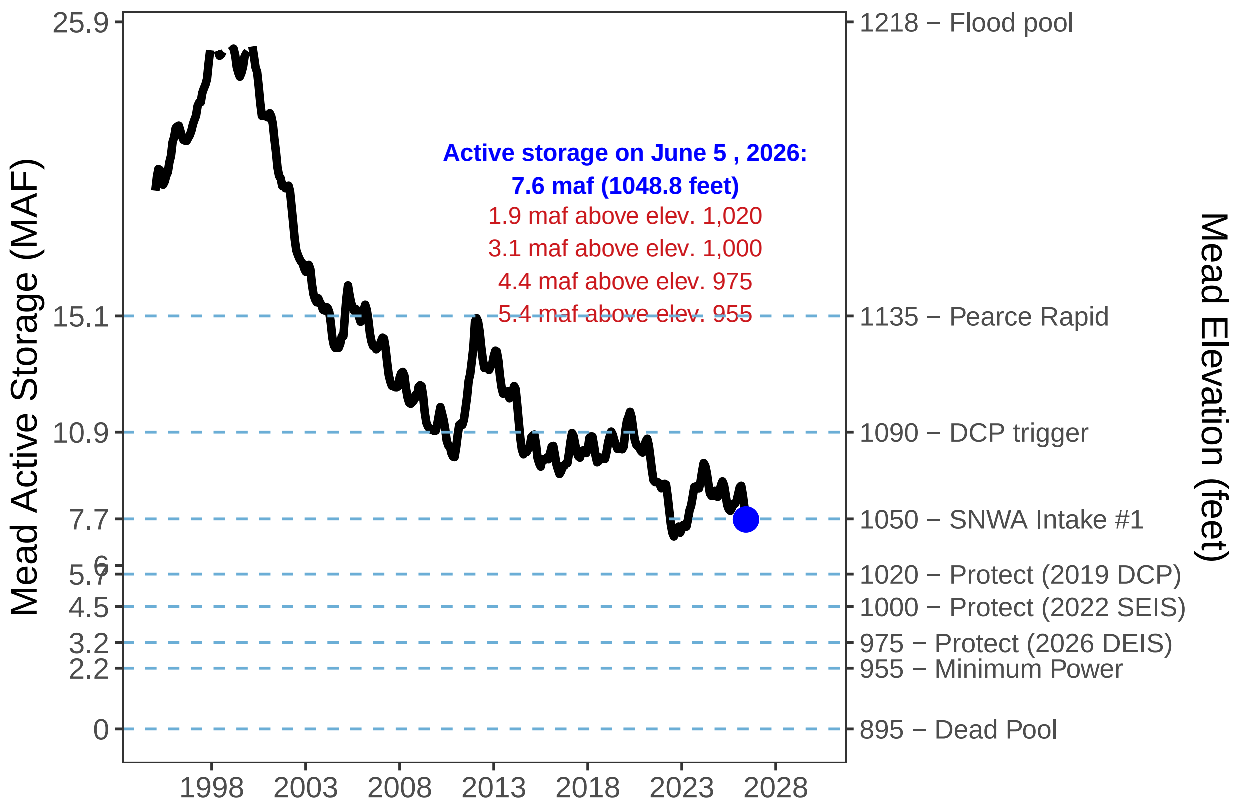

Figure 6B: Lake Powell Daily Inflow and Release Water Year to Date

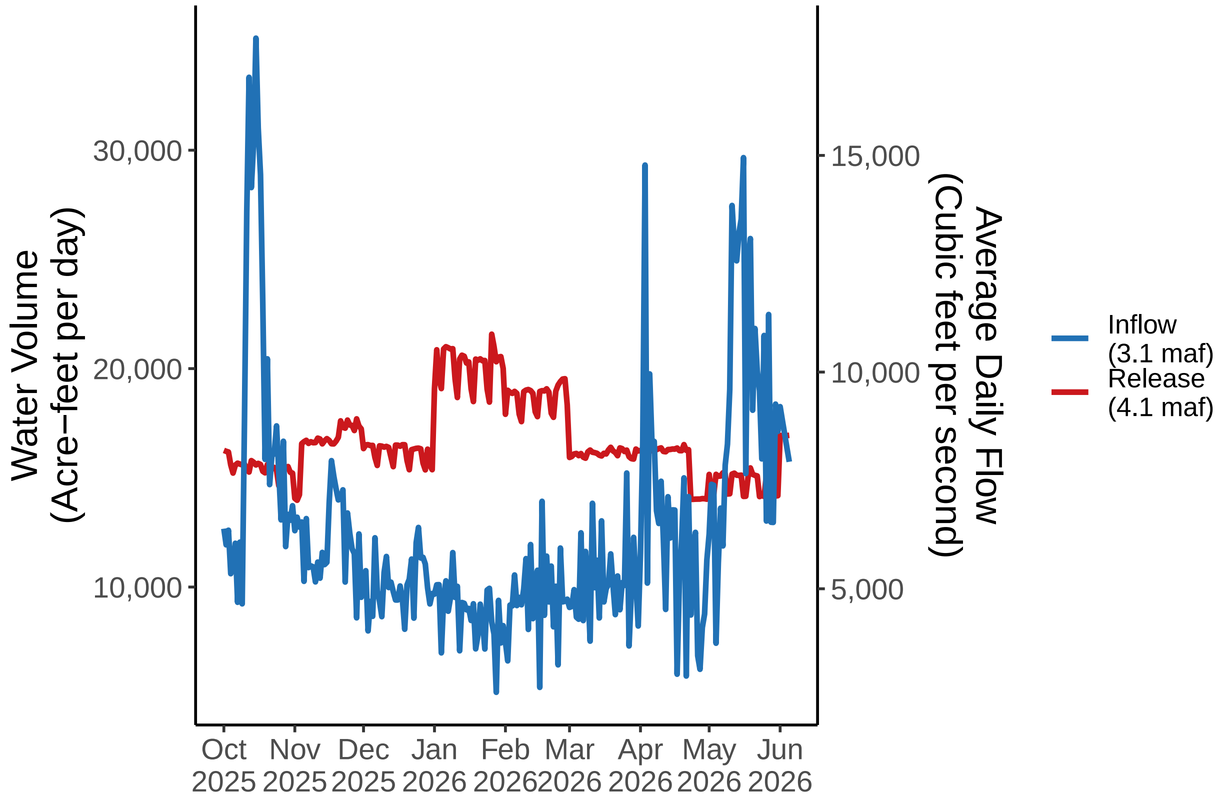

Figure 7: Frequency of Annual Lake Powell Release Volumes.

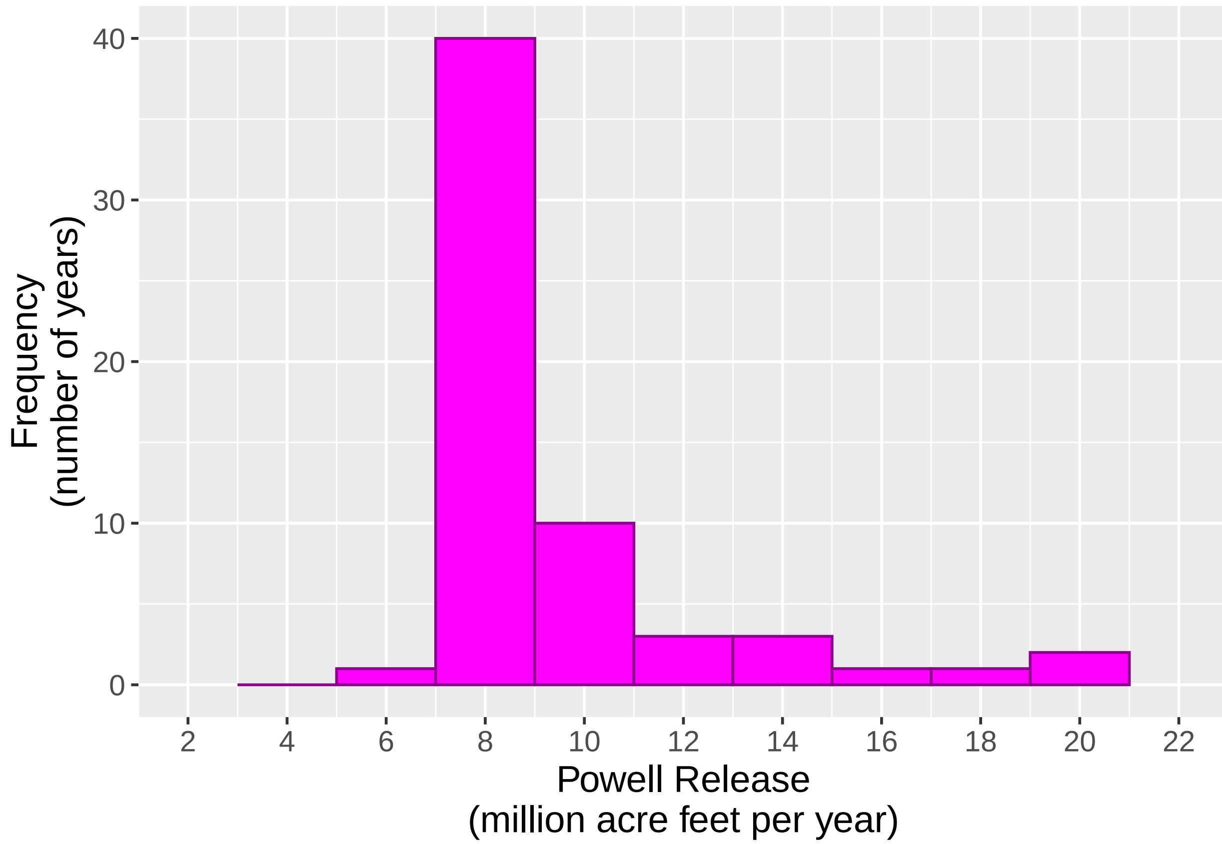

Figure 8: Frequency of Annual Lake Powell Inflow Volumes.

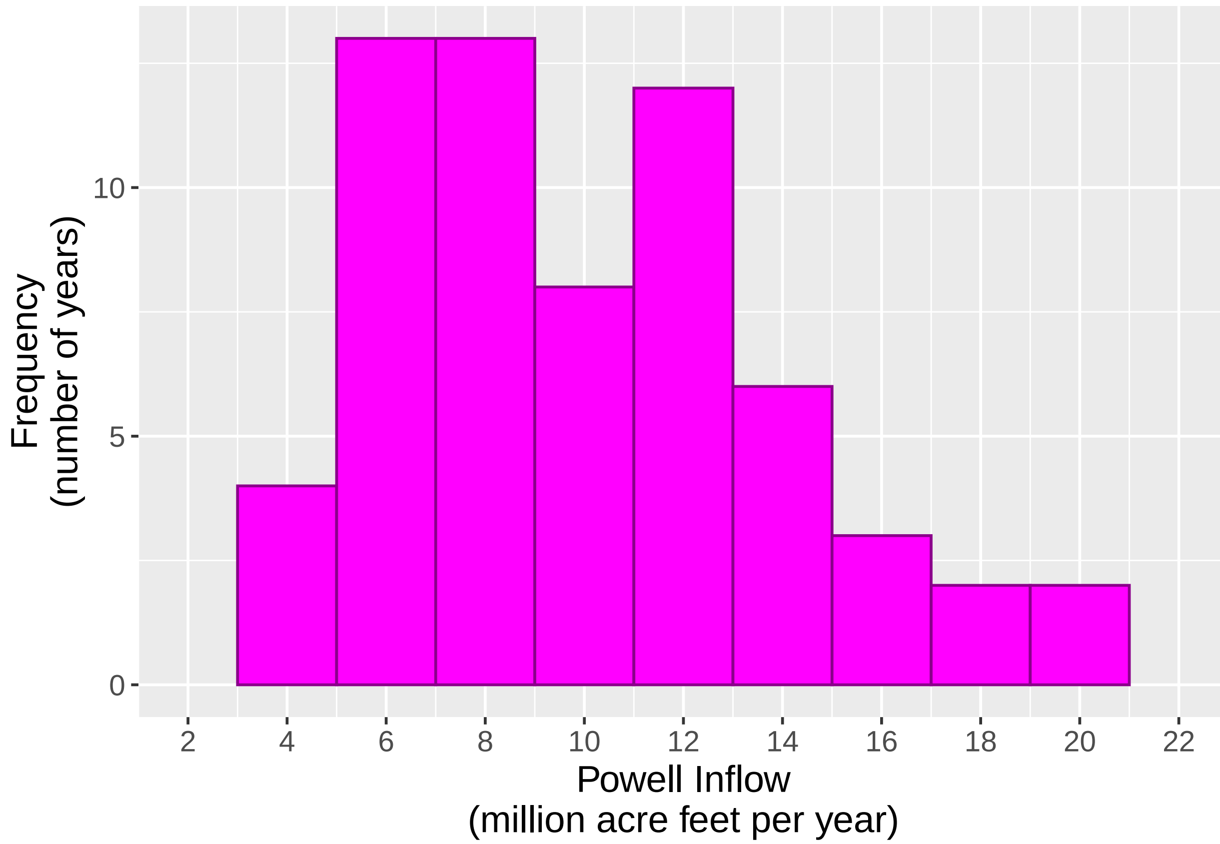

Figure 9: Lake Powell Inflows and Releases.

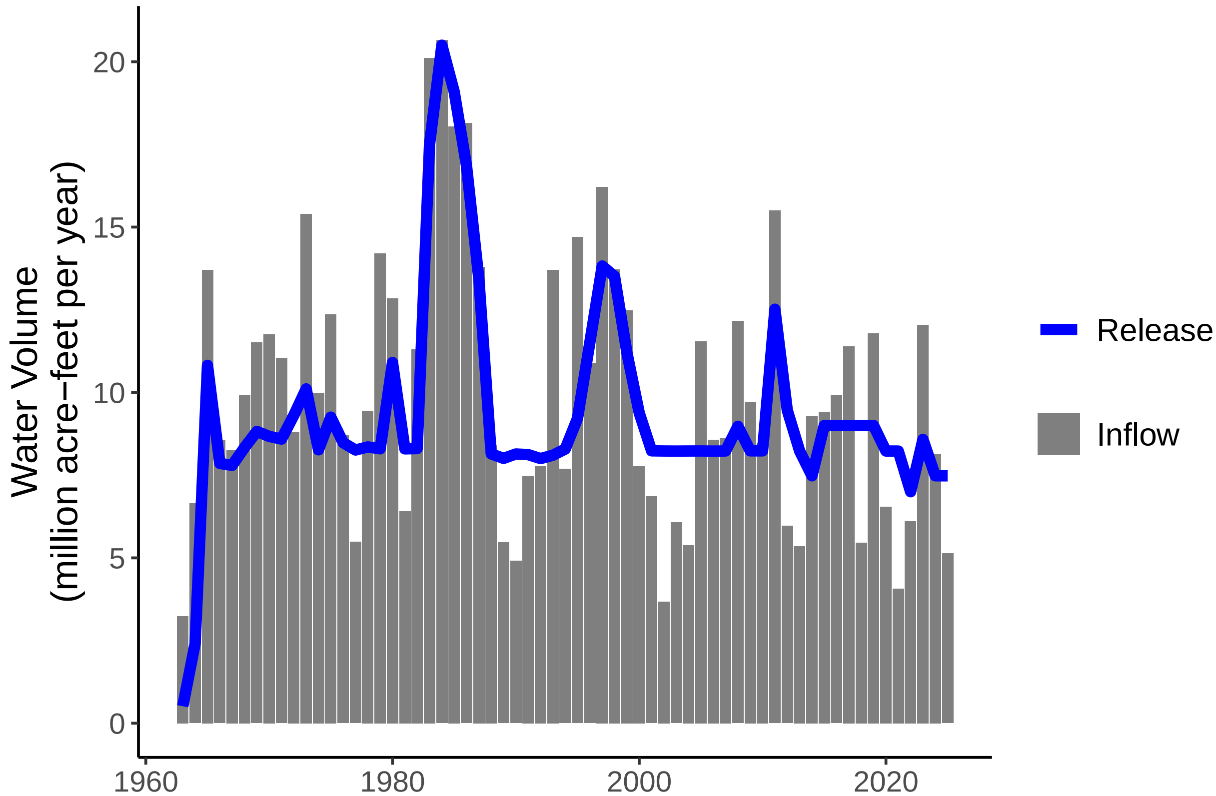

Figure 10: Annual Lake Powell Evaporation.